Meaning-Use Diagrams for Arithmetic Strategies

EPLE Automated Analysis

October 6, 2025

Contents

1 Introduction

This document presents automatically generated Meaning-Use Diagrams (MUDs) showing how arithmetic strategies elaborate upon each other through shared computational patterns.

Each diagram shows:

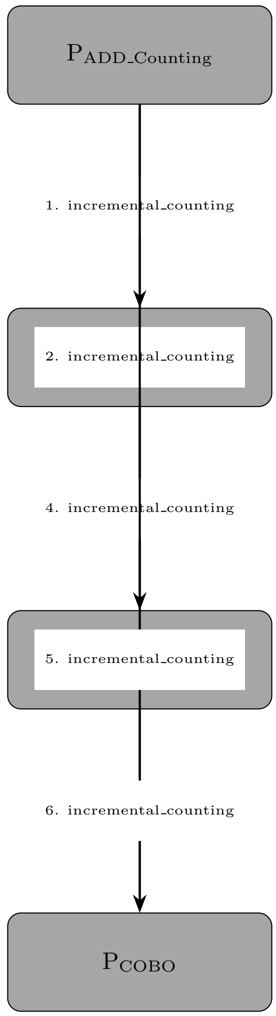
- **Strategy nodes** (gray rounded rectangles): Individual arithmetic strategies
- **Elaboration arrows**: Show how one strategy builds upon another
- **Pattern labels**: Computational patterns shared between strategies

2 Addition Strategies

2.1 Overview

This diagram shows 4 addition strategies and their 6 elaboration relationships based on shared incremental counting patterns.

2.2 Diagram



2.3 Interpretation

ADD_Counting serves as the foundational strategy, using basic incremental counting. This pattern is then elaborated in multiple ways:

- **ADD_Chunking**: Extends counting to work with chunks/groups
- **ADD_COBO**: Counting On from Bigger Operand
- **COBO**: Generic Counting On from Bigger Operand (cross-operation)

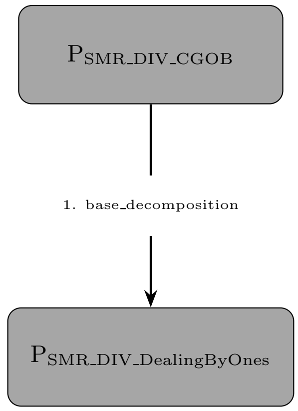
The transitive elaborations show that more complex strategies (ADD_COBO, COBO) also share the incremental counting pattern with intermediate strategies (ADD_Chunking).

3 Division Strategies

3.1 Overview

This diagram shows 2 division strategies with 1 elaboration relationship based on base decomposition.

3.2 Diagram



3.3 Interpretation

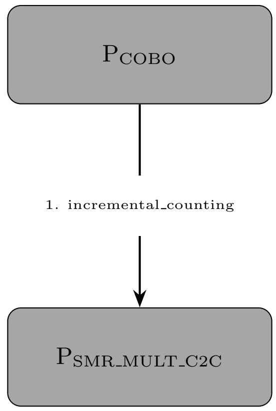
SMR_DIV_CGOB (Converting to Groups of Base) uses base decomposition as its core computational pattern. This elaborates to **SMR_DIV_DealingByOnes**, which also employs base decomposition but in the context of distributing units one at a time.

4 Cross-Operation Patterns

4.1 Overview

This diagram shows 2 strategies from different operations that share computational patterns.

4.2 Diagram



4.3 Interpretation

This demonstrates an inter-categorial elaboration: the subtraction strategy **COBO** shares the incremental counting pattern with the multiplication strategy **SMR_MULT_C2C** (Count by Composites). This reveals how conceptual metaphors transcend operational boundaries.

5 Conclusion

These diagrams reveal the hidden structure of mathematical cognition through automated analysis of actual computational implementations. The patterns identified (incremental counting, base decomposition) serve as the pragmatic infrastructure enabling algorithmic elaboration across and within arithmetic operations.